

Diagonalization of a Matrix (矩阵的对角化)

Lecture 22

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Diagonalization of a Matrix

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Introduction

We start right off with the one essential computation. It is perfectly simple and will be used in every section of this chapter. The eigenvectors diagonalize a matrix:

Theorem

Suppose the n by n matrix A has n linearly independent eigenvectors. If these eigenvectors are the columns of a matrix S , then $S^{-1}AS$ is a diagonal matrix Λ . The eigenvalues of A are on the diagonal of Λ :

$$S^{-1}AS = \Lambda = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix}$$

Diagonalization of a Matrix

We call S the “eigenvector matrix” and Λ the “eigenvalue matrix”.

There are four remarks before giving any examples or applications.

- 1: Any matrix with distinct eigenvalues can be diagonalized. If the matrix A has no repeated eigenvalues—the numbers $\lambda_1, \lambda_2, \dots, \lambda_n$ are distinct—then its n eigenvectors are automatically independent. Therefore any matrix with distinct eigenvalues can be diagonalized.
- 2: The diagonalizing matrix is not unique. An eigenvector x can be multiplied by a constant, and remains an eigenvector.

Remarks

- 3: Other matrices S will not produce a diagonal Λ .
- 4: Not all matrices possess n linearly independent eigenvectors, so **not all matrices are diagonalizable**. The standard example of a “defective matrix” is

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

$\lambda = 0$ is a double eigenvalue—its algebraic multiplicity is 2. But the geometric multiplicity is 1—there is only independent eigenvector. We can not construct S .

Diagonalizability and Invertibility

There is **no** connection between diagonalizability and invertibility. Diagonalization can fail **only if** there are **repeated** eigenvalues. The problem is the shortage of eigenvectors—which are needed for S . That needs to be emphasized:

Proposition

Diagonalizability of A depends on enough eigenvectors. Invertibility of A depends on nonzero eigenvalues.

Algebraic multiplicity and geometric multiplicity.

Theorem

Theorem

If eigenvectors $x_1, x_2, \dots, x_k$ correspond to different eigenvalues $\lambda_1, \dots, \lambda_k$, then those eigenvectors are linearly independent.

Proof.

Suppose first that $k = 2$, and that some combination of x_1 and x_2 produces zero: $c_1x_1 + c_2x_2 = 0$. Multiplying by A , we find $c_1\lambda_1x_1 + c_2\lambda_2x_2 = 0$.

Subtracting λ_2 times the previous equation, the vector x_2 disappears:

$c_1(\lambda_1 - \lambda_2)x_1 = 0$. Since $\lambda_1 \neq \lambda_2$ and $x_1 \neq 0$, we are forced into $c_1 = 0$.

Similarly $c_2 = 0$, and the two vectors are independent; only the trivial combination gives zero. The same argument extends to any number of vectors. □

Examples of Diagonalization

The main point of this section is $S^{-1}AS = \Lambda$. The eigenvector matrix S converts A into its eigenvalue matrix Λ (diagonal). We see this for projections and rotations.

Example 1. The projection $A = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$ has eigenvalue matrix

$$\Lambda = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Example 2

Example 2. The eigenvalues and eigenvectors of a rotation matrix. 90° rotation:

$$K = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

How can a vector be rotated and still have its direction unchanged?

Imaginary numbers? Complex numbers are needed even for real matrices! See section 5.5 for more!

Powers and Products: A^k and AB

The eigenvalues of A^2 are exactly $\lambda_1^2, \dots, \lambda_n^2$, and every eigenvector of A is also an eigenvector of A^2 .

Theorem

The eigenvalues of A^k are $\lambda_1^k, \dots, \lambda_n^k$, and each eigenvector of A is still an eigenvector of A^k . When S diagonalizes A , it also diagonalizes A^k :

$$A^k = S^{-1} A^k S$$

Each S^{-1} cancels an S , except for the first S^{-1} and the last S .

If A is invertible, this rule also applies to its inverse (the power $k = -1$). The eigenvalues of A^{-1} are $1/\lambda_i$. That can be seen without diagonalizing:

$$\text{if } Ax = \lambda x \text{ then } x = \lambda A^{-1}x \text{ and } \frac{1}{\lambda}x = A^{-1}x.$$

Example 3

Example 3. If K is rotation through 90° , then K^2 is rotation through 180° (which means $-I$) and K^{-1} is rotation through -90° :

$$K = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, K^2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}, \text{ and } K^{-1} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

The eigenvalues of K are i and $-i$; their squares are -1 and -1 ; their reciprocals are $1/i = -i$ and $1/(-i) = i$. Then K^4 is a complete rotation through 360° .

$$K^4 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \text{ and also } \Lambda^4 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Notice also that complex numbers are needed even for real matrices.

Product

- For a product of two matrices, we can ask about the eigenvalues of AB —but we won't get a good answer.
- In general, if A has an eigenvalue λ and B has an eigenvalue μ , AB does not have $\lambda\mu$ as its eigenvalue. For instance, we have two matrices with zero eigenvalues, while AB has $\lambda = 1$:

$$AB = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

The eigenvectors of this A and B are completely different, which is typical.

- Similarly, the eigenvalues of $A + B$ generally have nothing to do with $\lambda + \mu$.

Theorem

If the eigenvector is the same for A and B , then the eigenvalues multiply and AB has the eigenvalue $\mu\lambda$. But there is something more important.

There is an easy way to recognize when A and B share a full set of eigenvectors, and that is a key question in quantum mechanics:

Theorem

Diagonalizable matrices share the same eigenvector matrix S if and only if $AB = BA$.

Proof

Proof. If the same S diagonalizes both $A = S\Lambda_1 S^{-1}$ and $B = S\Lambda_2 S^{-1}$, we can multiply in either order:

$$AB = S\Lambda_1 S^{-1} S\Lambda_2 S^{-1} = S\Lambda_1 \Lambda_2 S^{-1} \text{ and } BA = S\Lambda_2 S^{-1} S\Lambda_1 S^{-1} = S\Lambda_2 \Lambda_1 S^{-1}.$$

Since $\Lambda_1 \Lambda_2 = \Lambda_2 \Lambda_1$ (diagonal matrices always commute) we have $AB = BA$.

Proof

In the opposite direction, suppose $AB = BA$. Starting from $Ax = \lambda x$, we have

$$ABx = BAx = B\lambda x = B(\lambda x) = \lambda Bx.$$

Thus x and Bx are both eigenvectors of A , sharing the same λ . If we assume for convenience that the eigenvalues of A are distinct—the eigenspaces are all one-dimensional—then Bx must be a multiple of x . In other words, x is an eigenvector of B as well as A . The proof with repeated eigenvalues is a little longer (left as an exercise).

Heisenberg's uncertainty principle

Heisenberg's uncertainty principle comes from non-commuting matrices, like position P and momentum Q . Position is symmetric, momentum is skew-symmetric, and together they satisfy:

$$QP - PQ = I.$$

The uncertainty principle follows directly from the Schwarz inequality $(Qx)^T(Px) \leq \|Qx\| \|Px\|$ of Section 3.2:

$$\|x\|^2 = x^T x = x^T (QP - PQ)x \leq 2\|Qx\| \|Px\|$$

The product of $\|Qx\|/\|x\|$ and $\|Px\|/\|x\|$ —momentum and position errors, when the wave function is x —is at least $\frac{1}{2}$. It is impossible to get both errors small, because when you try to measure the position of a particle you change its momentum.

Final Note

At the end we come back to $A = S\Lambda S^{-1}$.

- That factorization is particularly suited to take powers of A , and the simplest case A^2 makes the point.
- The LU factorization is hopeless when squared, but $S\Lambda S^{-1}$ is perfect. The square is $S\Lambda^2 S^{-1}$, and the eigenvectors are unchanged.
- By following those eigenvectors we will solve difference equations and differential equations.

Homework Assignment 22

5.2: 7, 8, 12, 13, 19, 23, 32, 40.